Heart Disease Risk Prediction

Deliverable 3: Experiment Tracking with MLflow

Dataset: UCI Heart Disease (Cleveland, Hungary, Switzerland, VA Long Beach)
Models: Logistic Regression, Random Forest
Tracking: MLflow 3.17.1 (local SQLite backend, Model Registry)

1. Purpose of experiment tracking

In Deliverable 2, model results existed only as notebook outputs: re-running a cell overwrote them, and there was no lasting record of which settings, data or code produced a given score. Deliverable 3 integrates **MLflow** so that every training run is recorded permanently and can be compared, traced and reproduced. The chosen model is then registered in the **MLflow Model Registry**, so later stages (API, container, CI/CD) can load it by name instead of by file path.

2. What is logged for each run

MLflow concept	What this project logs
Parameters	Model hyperparameters (e.g. C, l1_ratio, max_depth, n_estimators), random seed, number of CV folds, test-set size, feature-set choice, training row count
Metrics	Accuracy, precision, recall and ROC-AUC from 5-fold cross-validation: mean, standard deviation and the value for each fold (logged as a step series). Tuned runs also receive test-set metrics.
Artifacts	ROC curve and confusion matrix (out-of-fold predictions), feature-importance plot, per-fold results table, grid-search results table, and the full model pipeline (preprocessing + model) with its input/output signature
Tags & lineage	Stage (feature_selection, baseline, tuning, tuned), model type, git commit of the code, data fingerprint (digest) and the training dataset as a logged input
Model Registry	The selected model registered as *heart-disease-classifier* with the alias *@champion*

3. How the runs are organised

All runs belong to one experiment, **heart-disease-classification**, giving **52 runs** in total:

- **Feature-set test (2 runs):** keep the sparse columns ca, thal and slope with missing-value indicators, versus dropping them.
- **Baselines (2 runs):** Logistic Regression and Random Forest with default settings.
- **Tuning (2 parent runs):** one per model. Each parent contains a child run for every hyperparameter combination tried (8 for Logistic Regression, 36 for Random Forest) plus one fully logged *tuned* run.
- **Selection rule:** the model with the higher mean CV ROC-AUC is chosen, unless the difference is within one standard deviation, in which case the simpler, more interpretable Logistic Regression is preferred.

FIGURE 1

MLflow tracking server and experiment

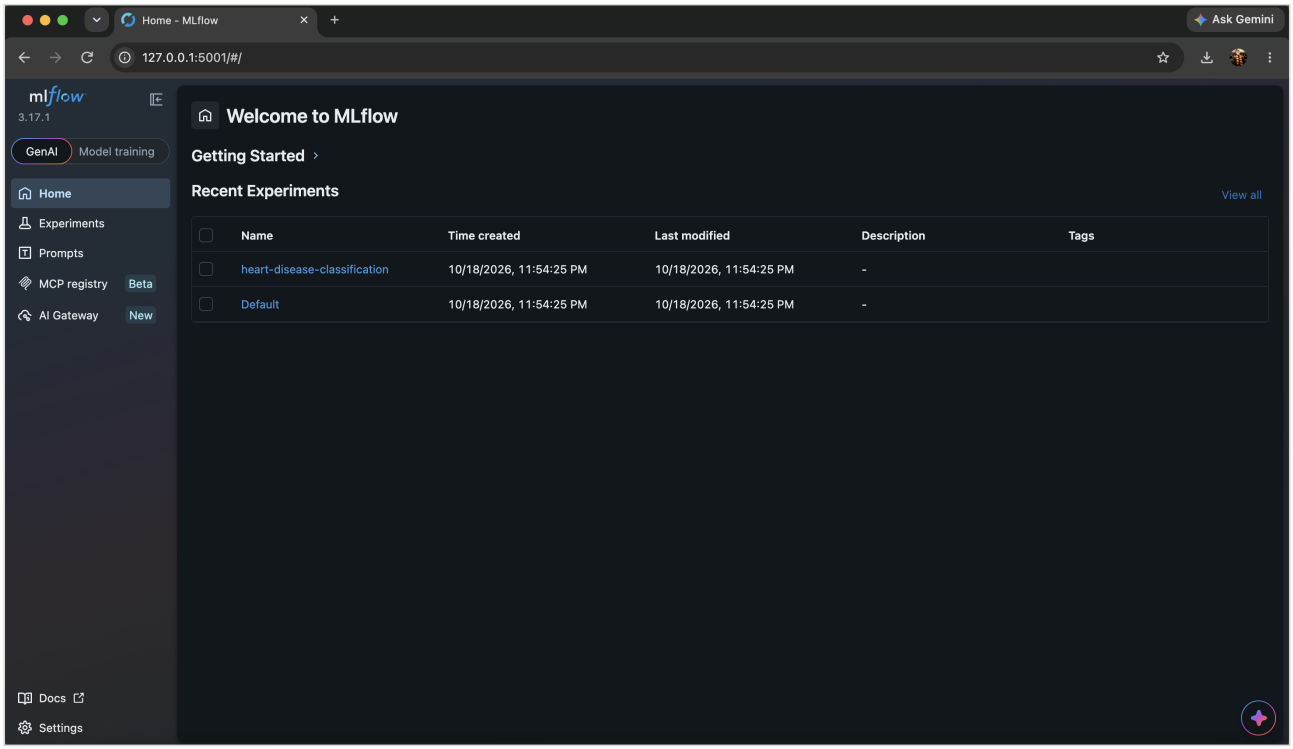


Figure 1. MLflow home page showing the project experiment.

The MLflow UI (version 3.17.1) running locally at *127.0.0.1:5001*. The **heart-disease-classification** experiment was created on 18 October 2026 at 11:54 PM, alongside MLflow's built-in *Default* experiment, which is unused.

- The tracking data is stored in a local SQLite database (*mlflow.db*) and the artifacts in *mlruns/*, both in the project folder and excluded from Git.
- Grouping all runs under one named experiment keeps this project's results separate and searchable.

FIGURE 2

All 52 runs in the experiment

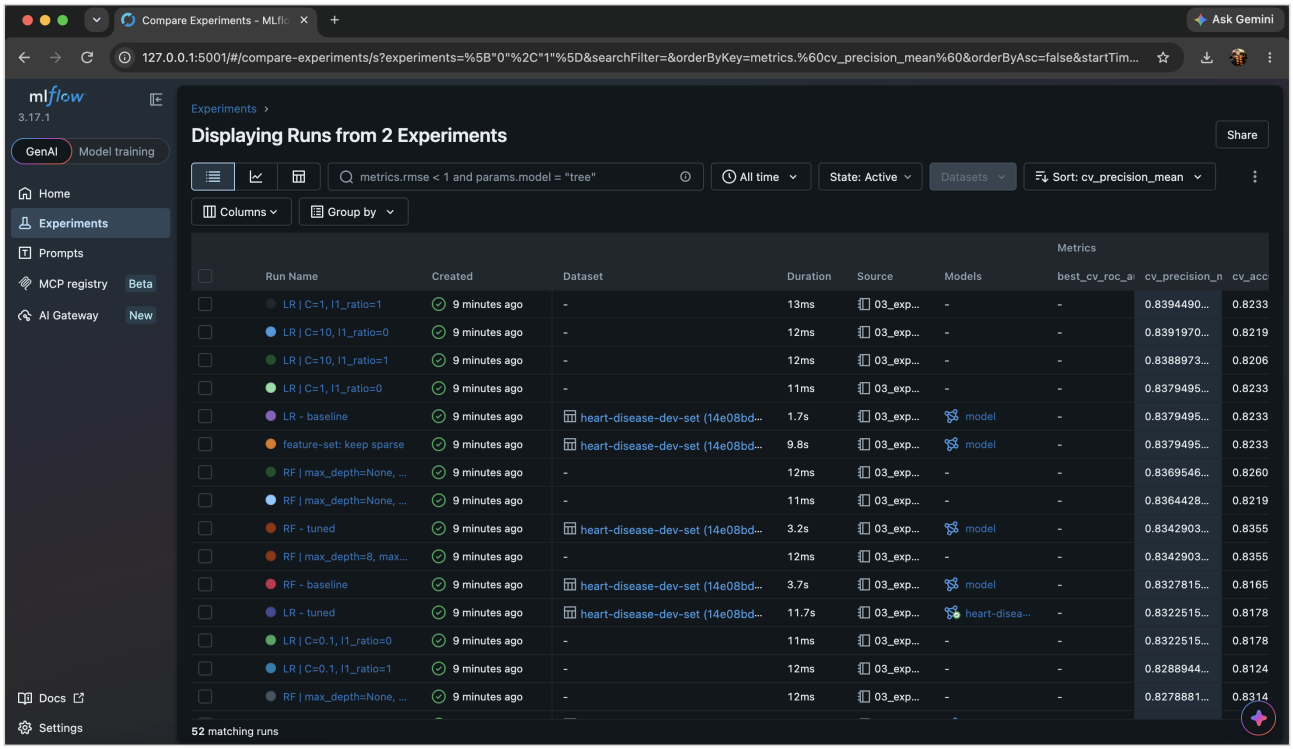


Figure 2. Run list with parameters, dataset lineage, logged models and metrics.

The run table lists **52 matching runs**, sorted by mean CV precision. Both the fully logged runs (e.g. *LR - baseline*, *RF - tuned*, *feature-set: keep sparse*) and the individual grid-search candidates (e.g. *LR | C=1, l1_ratio=1*, *RF | max_depth=8, …*) are recorded.

- **Dataset column:** full runs carry the logged input *heart-disease-dev-set (14e08bd…)*. The digest fingerprints the exact training data, so any result can be traced back to the data that produced it.
- **Models column:** each full run stores its trained pipeline; the *LR - tuned* run links to the registered model *heart-disease-classifier*.
- **Metrics columns:** cross-validated precision and accuracy are visible for every run, allowing direct comparison of all settings tried.
- The view header reads "Displaying Runs from 2 Experiments" because the empty Default experiment was also selected; all 52 runs belong to heart-disease-classification.

FIGURE 3

Automatic metric comparison charts

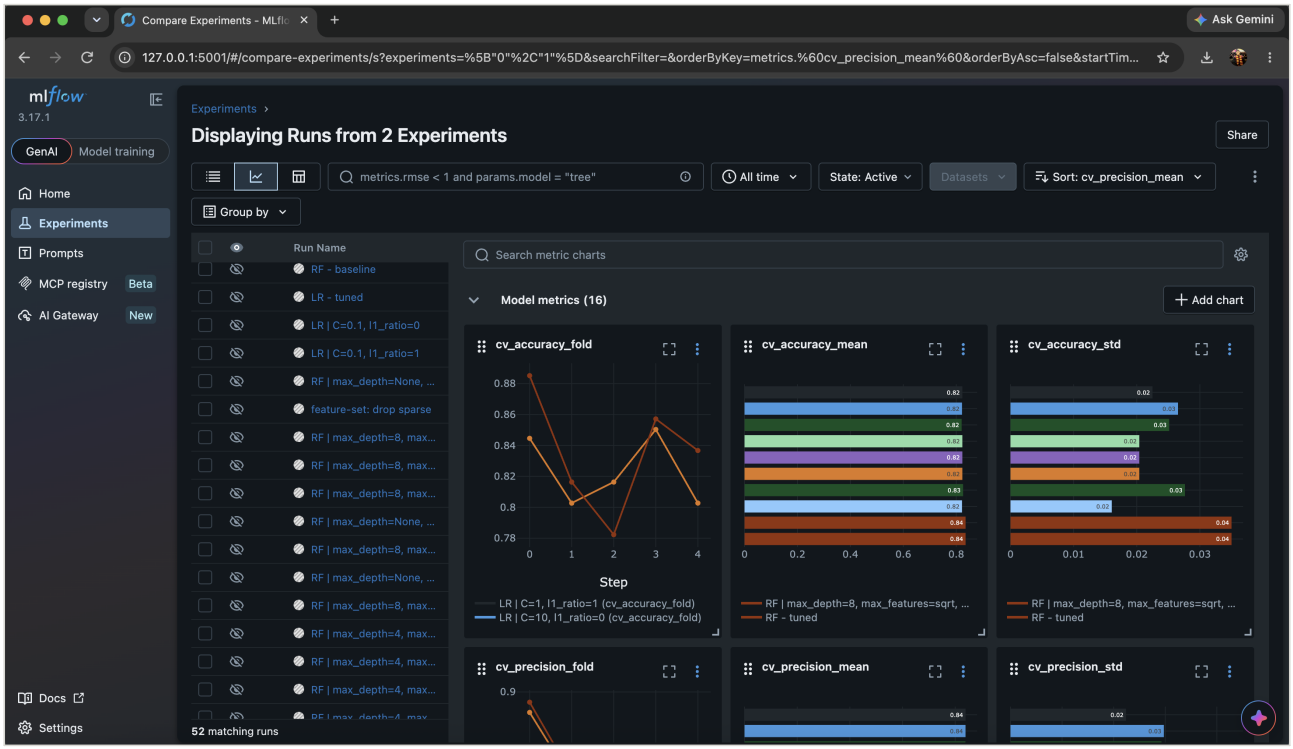


Figure 3. Metric charts: per-fold accuracy, mean accuracy and accuracy standard deviation.

MLflow's chart view generates **16 model-metric charts** directly from the logged metrics, without any plotting code.

- **cv_accuracy_fold:** accuracy for each of the 5 cross-validation folds (steps 0 to 4) for two Logistic Regression candidates. Fold scores range from roughly 0.78 to 0.88, which shows why results are reported as mean ± standard deviation rather than from a single split.
- **cv_accuracy_mean** and **cv_accuracy_std:** side-by-side comparison of runs. Mean accuracies lie around 0.82 to 0.84, with standard deviations between about 0.02 and 0.04.
- The precision charts begin below; equivalent charts exist for recall and ROC-AUC.

FIGURE 4

Model Registry: the champion model

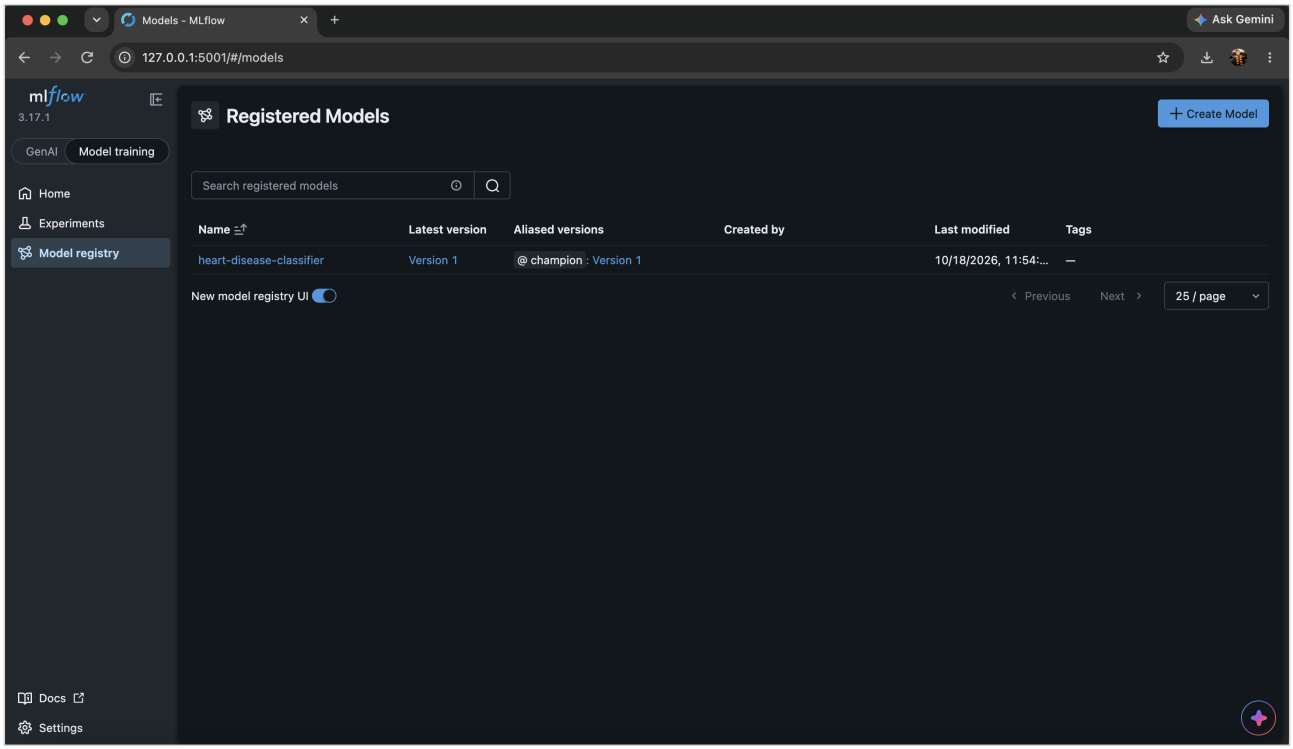


Figure 4. Registered model heart-disease-classifier with the @champion alias.

The Model Registry contains **heart-disease-classifier**, with **Version 1** as the latest version and the alias **@champion** pointing to Version 1.

- As shown in Figure 2, this version was registered from the *LR - tuned* run: the tuned **Logistic Regression** was selected under the rule above.
- Downstream components load the model as *models:/heart-disease-classifier@champion*. When a better model is registered later, moving the alias to the new version changes the served model without any code change.
- Each registry version links back to its run, so the parameters, metrics, data fingerprint and git commit behind the deployed model are always traceable.

4. Summary

MLflow experiment tracking is integrated into the training workflow. The screenshots demonstrate:

- **Parameters, metrics, artifacts and plots** are logged for all runs: 52 runs covering the feature-set test, both baselines, every hyperparameter combination and both tuned models.
- **Lineage and reproducibility:** each full run records the training dataset fingerprint and the git commit of the code that produced it.
- **Comparison:** runs can be filtered, sorted and charted in the UI, and queried from code.
- **Model management:** the selected model, the tuned Logistic Regression, is registered as *heart-disease-classifier* Version 1 with the *@champion* alias.

Next step

Deliverable 4 (Model Packaging & Reproducibility) will serve the champion model through a REST API that loads *models:/heart-disease-classifier@champion* directly from the registry.